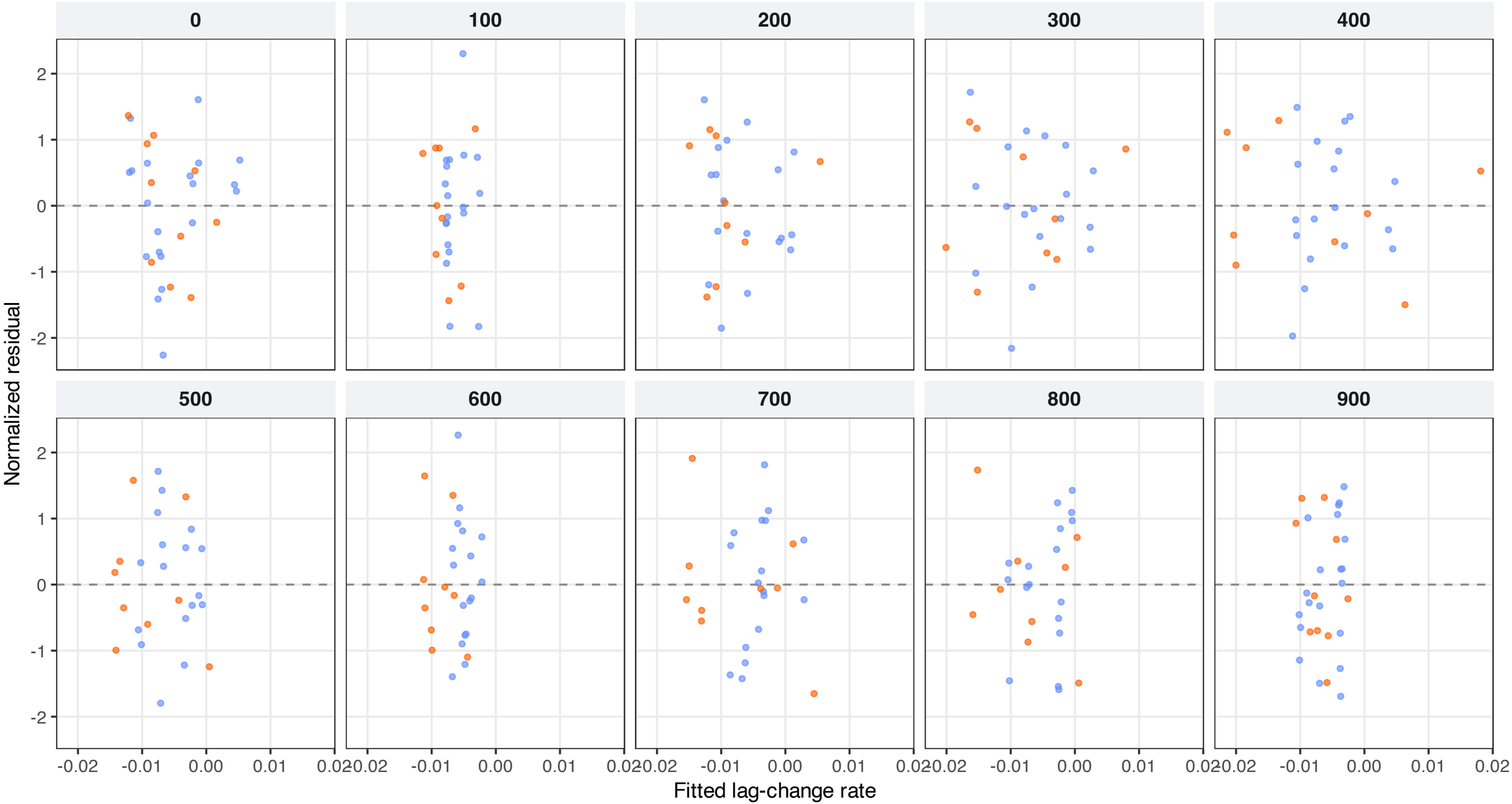
Observed lag change · Holocene normalized residuals

0 100 200 300 400

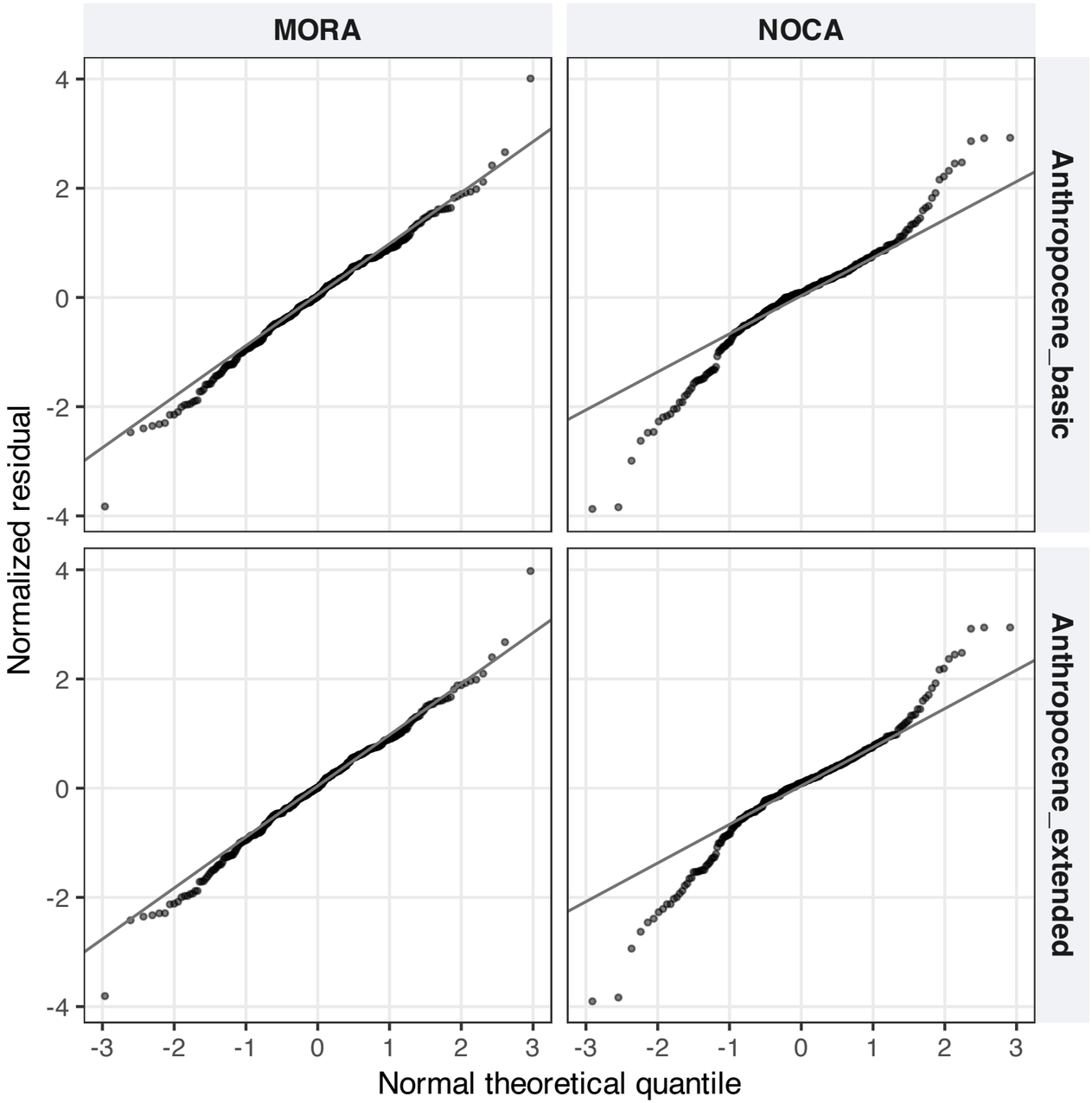
500 600 700 800 900

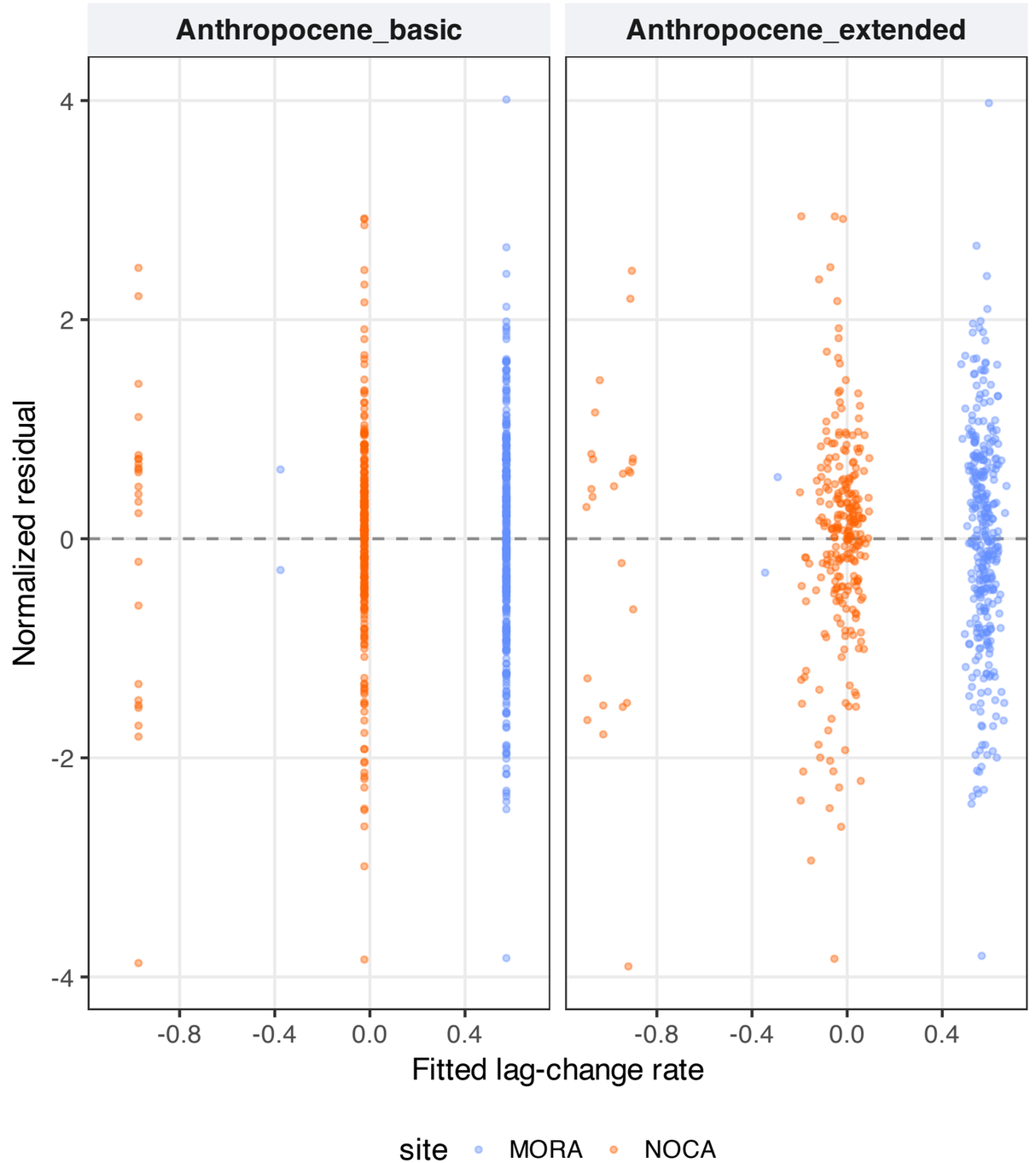
Normalized residual

Normal theoretical quantile

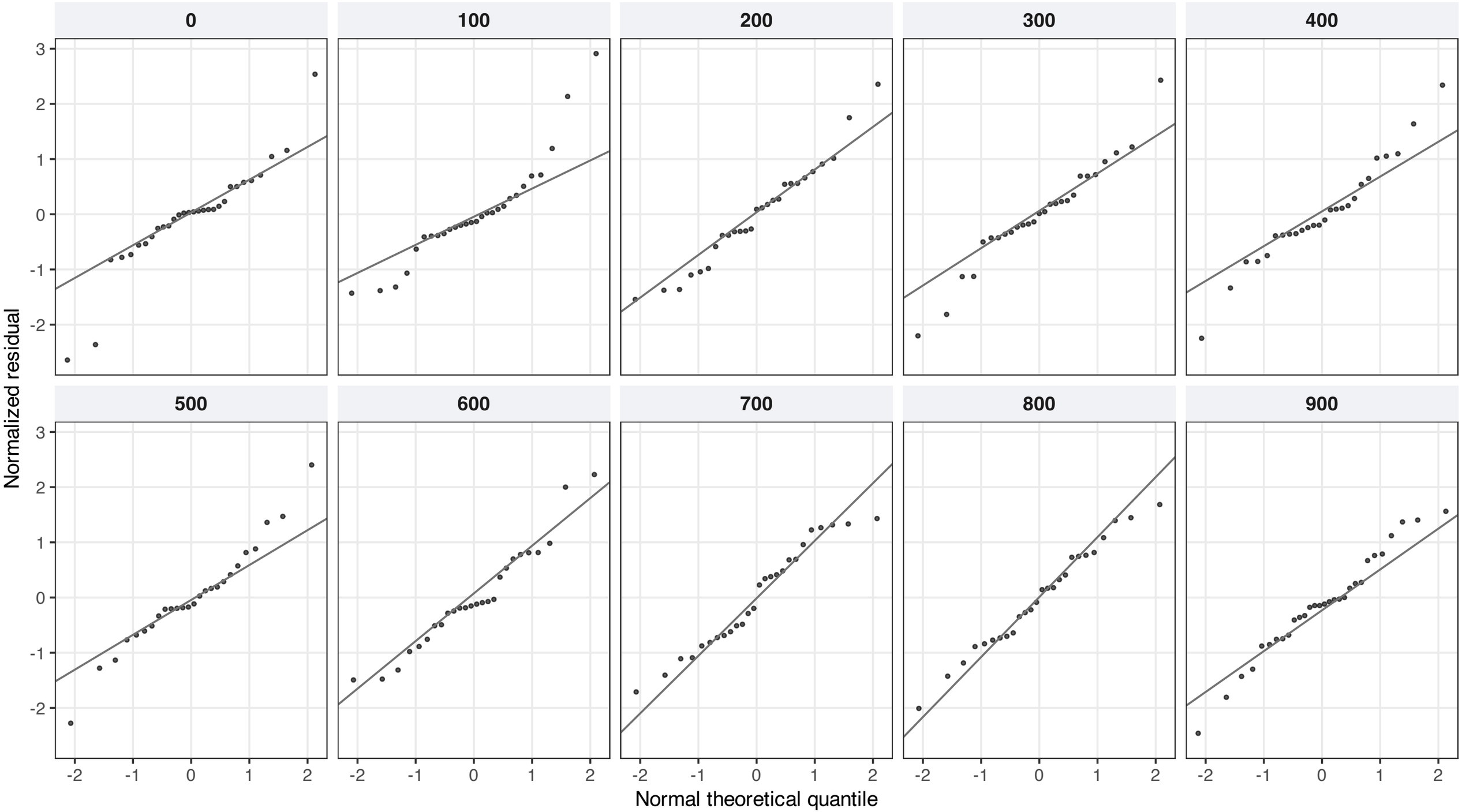

0
100
200
300
400
500
600
700
800
900
2
1
0
-1
-2
-0.02
-0.01
0.00
0.01
0.02
Normalized residual
Fitted lag-change rate
site
MORA
NOCA

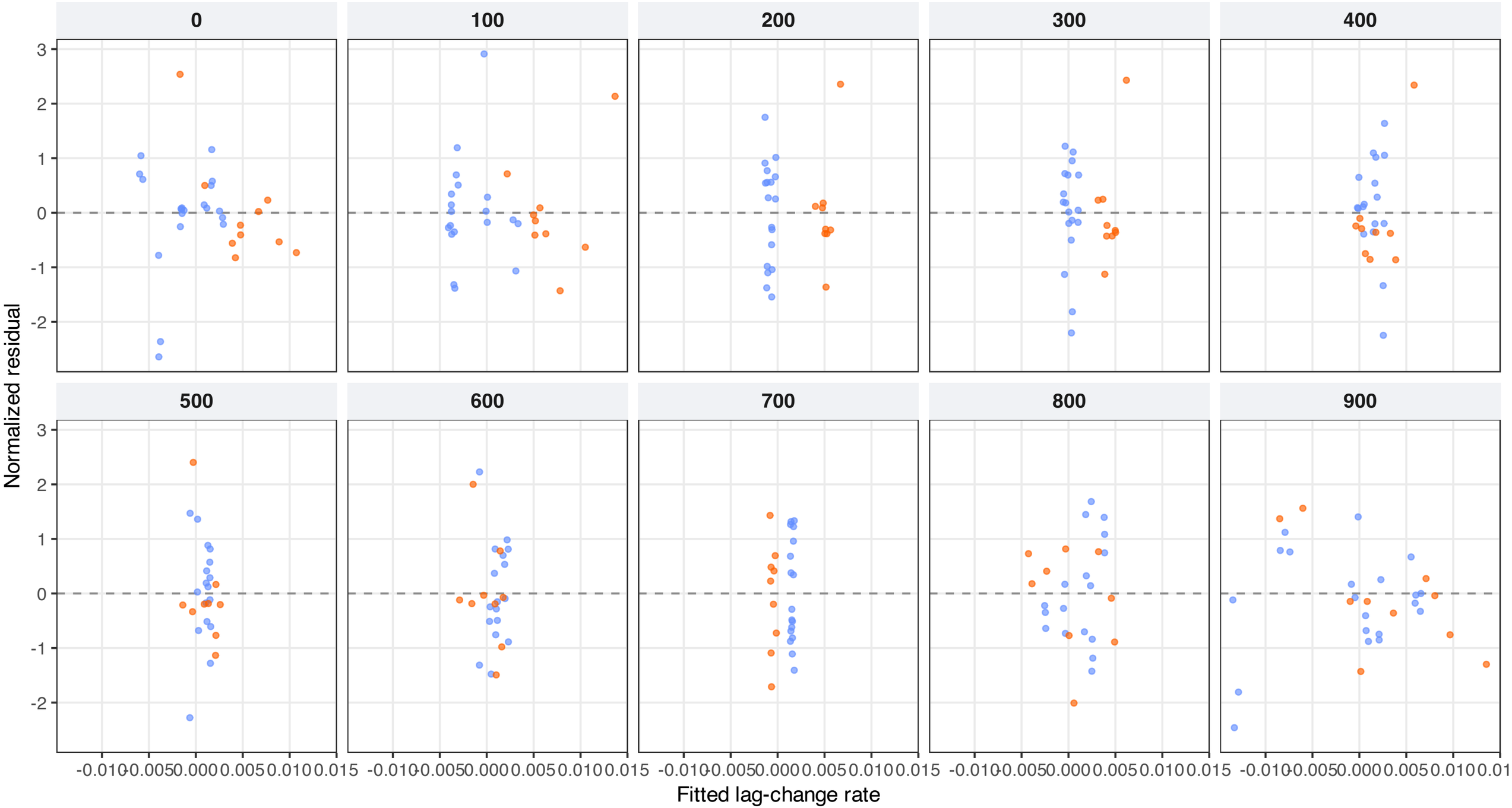
Observed lag change · modern models



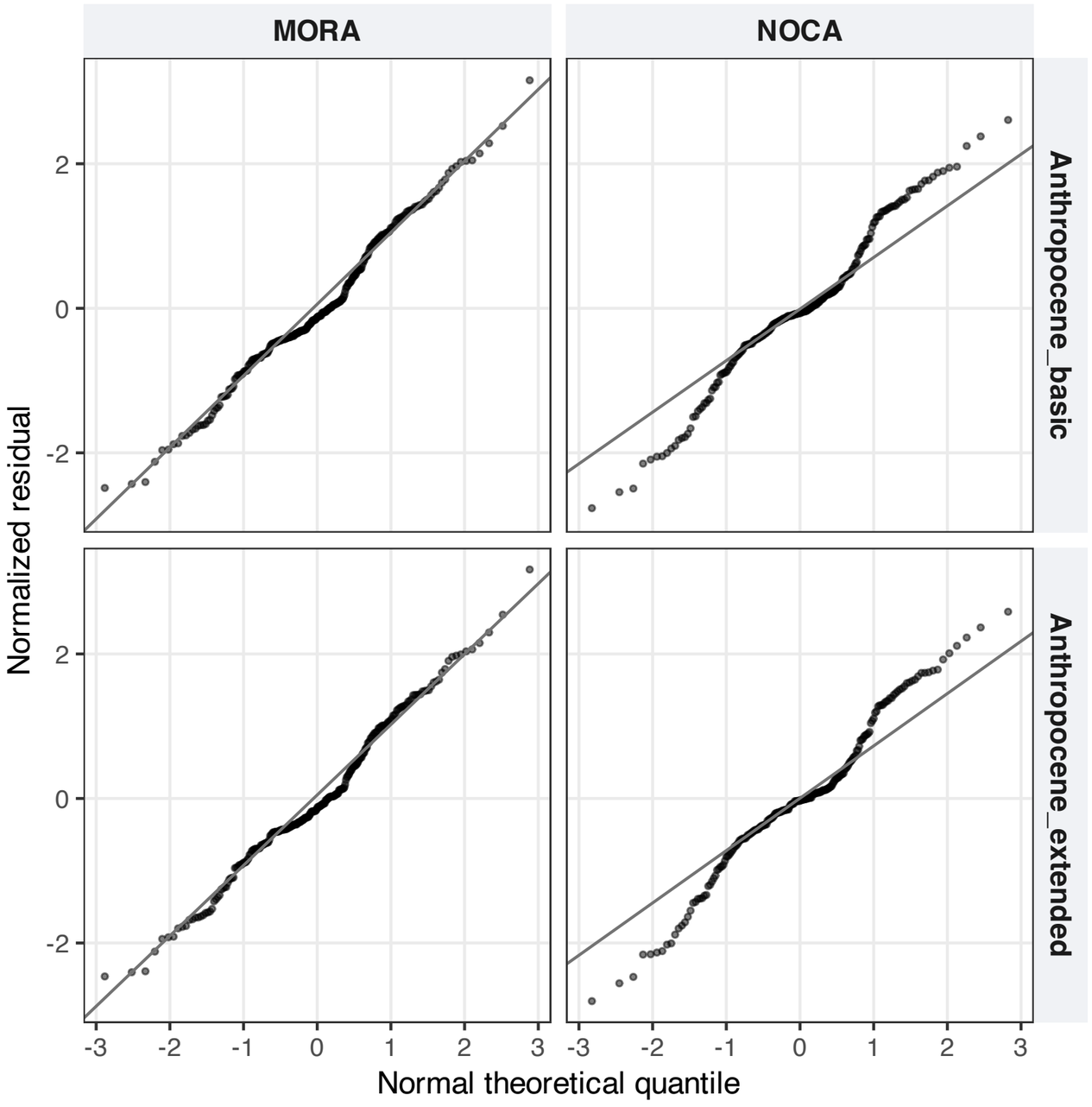


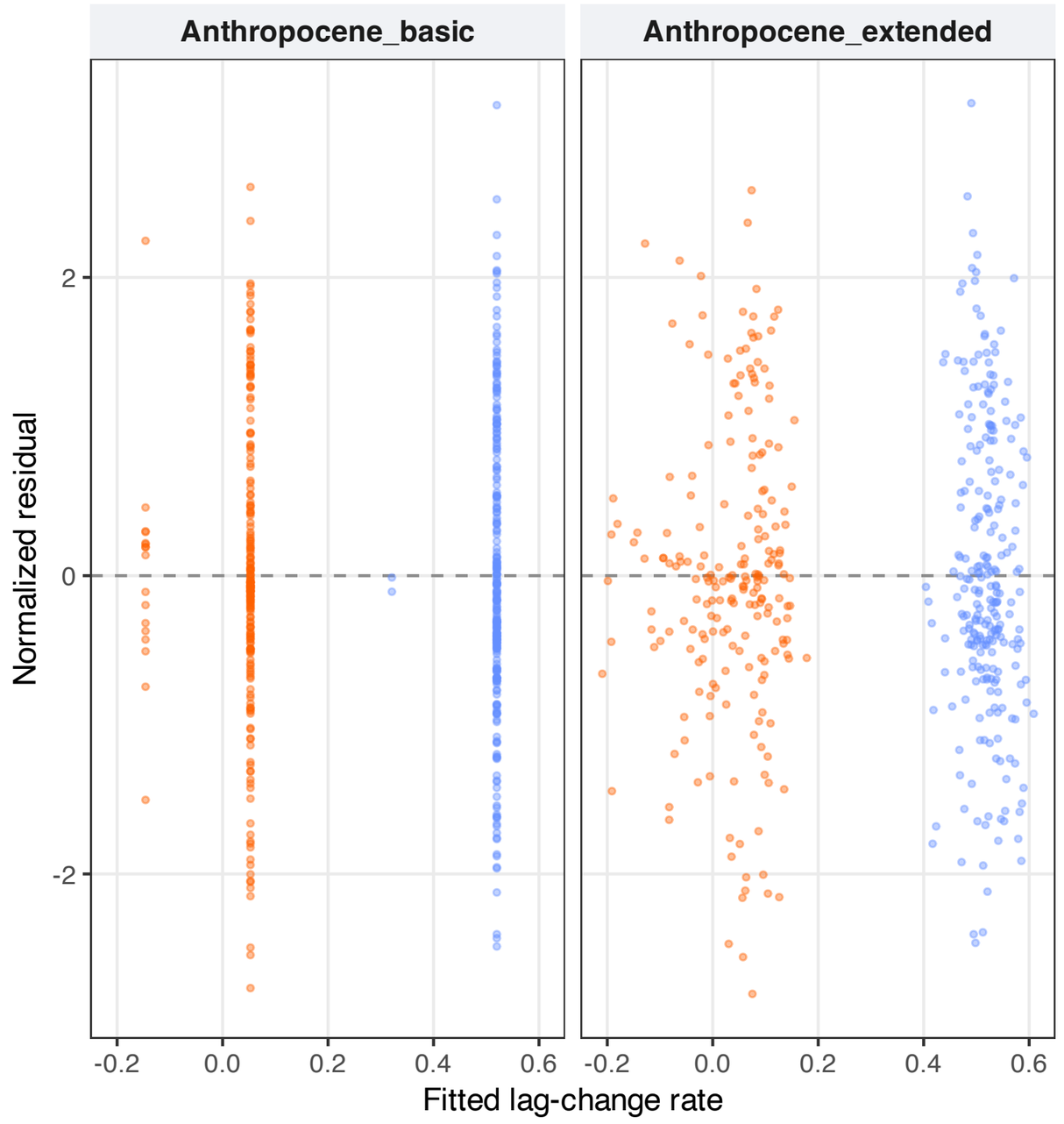
Pace-corrected lag change · Holocene normalized residuals




0
100
200
300
400
500
600
700
800
900
3
2
1
0
-1
-2
Normalized residual
-0.010
-0.005
0.000
0.005
0.010
0.015
Fitted lag-change rate
site
MORA
NOCA

Pace-corrected lag change · modern models





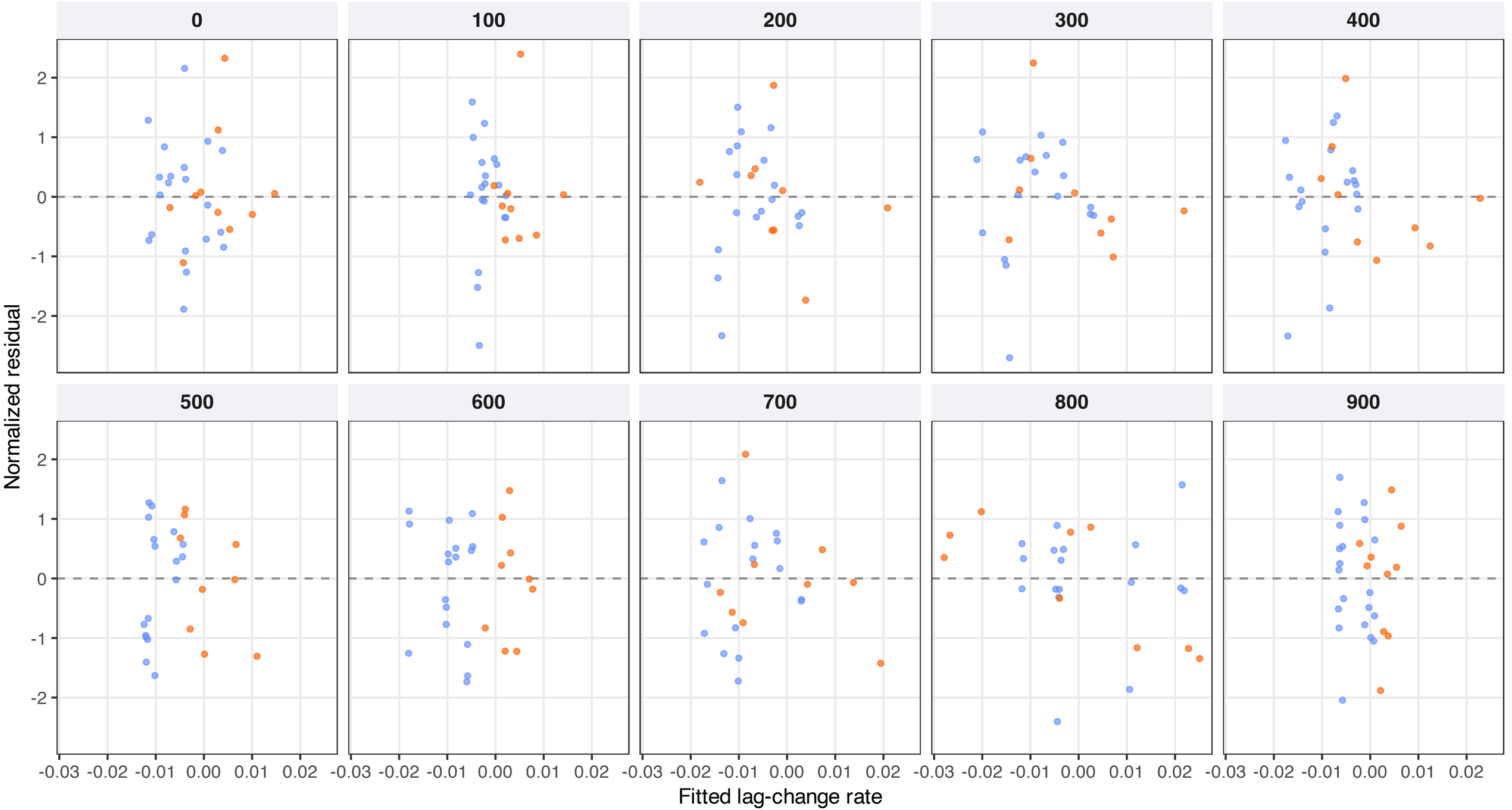
Direction-corrected lag change · Holocene normalized residuals

0 100 200 300 400

500 600 700 800 900

Normalized residual

Normal theoretical quantile


0
100
200
300
400
500
600
700
800
900
Normalized residual
2
1
0
-1
-2
-0.03 -0.02 -0.01 0.00 0.01 0.02
Fitted lag-change rate
site
MORA
NOCA

Direction-corrected lag change · modern models

